

RT-ZS-BZ-*

Onboard noise module

user's manual

Document version: V2.0



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1. product description

1.1 product description

RT-ZS-BZ-* onboard noise module, mainly in accordance with national standard GB/T 3785.1-2010 "Electrical Acoustic Sound Level Meter

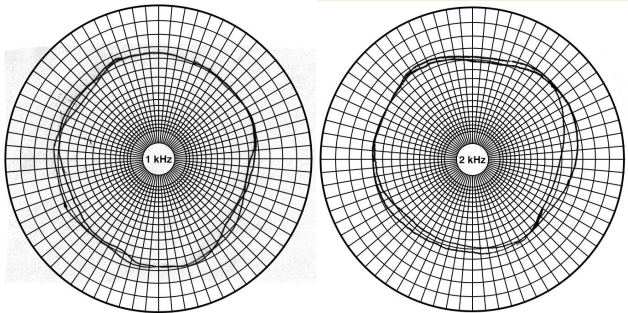
Part 1: Specification and International Standard IEC 61672-1:2002 Design requirements for Class 2 X-class sound level meters, mainly for environmental noise, traffic noise, workplace noise, construction noise and social life noise Real-time measurement of noise on the spot. With this module, customers no longer have to worry about complex noise signal processing, and can focus on areas of expertise and create value for customers faster.

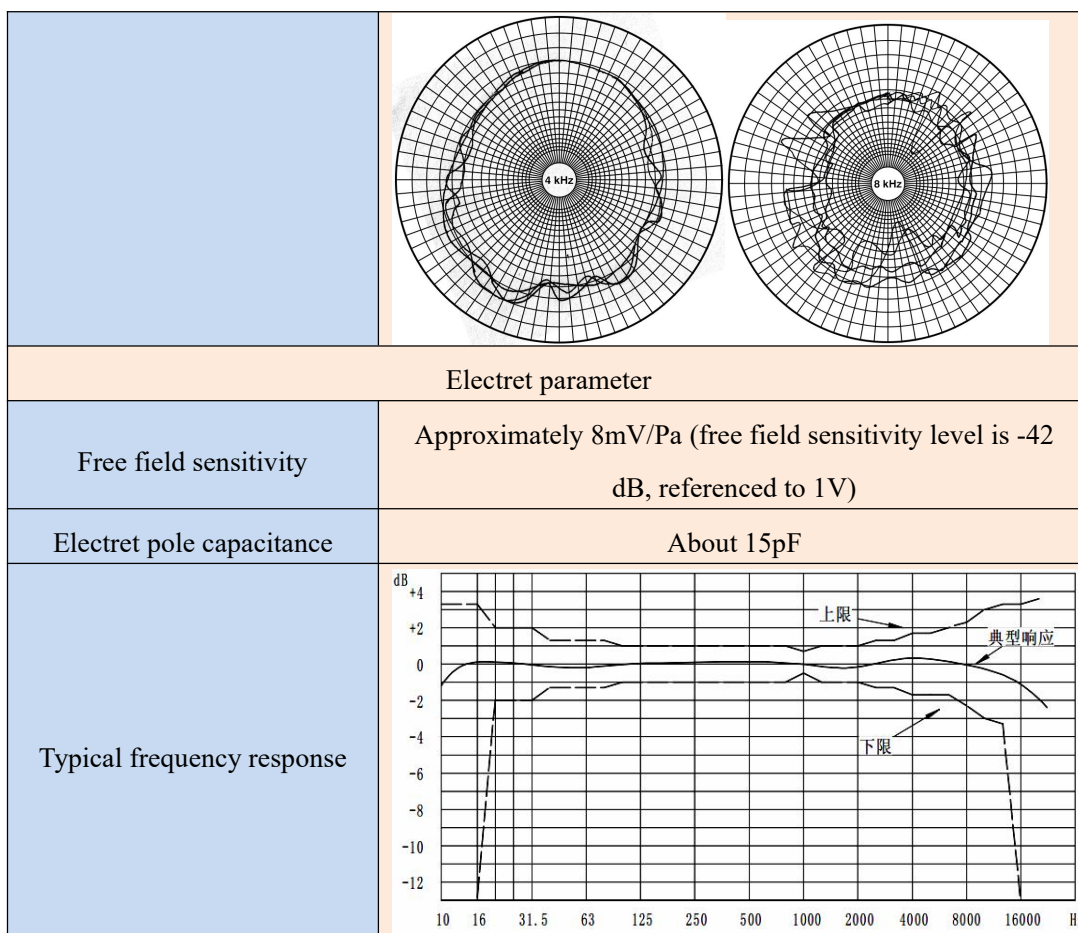
1.2 Features

1. PCB onboard installation method.
2. 30~120dBA wide range, 20~12.5KHz wide frequency measurement.
3. High-performance pre-polarized back-electrode electret condenser microphone with wide dynamic range and stable performance.
4. Output interface TTL or RS-485 is optional at the factory.
5. It has two modes of slow and fast measurement to meet the requirements of different customers.
6. Power supply 4.5V~5.5VDC, 10~28VDC optional.

1.3 Main Specifications

Operating Voltage	4.5~5.5V (default) 10~28V (Optional)	
Power consumption	18.9mA@5V 31.0mA@12V 27.8mA@24V	
Transmitter circuit operating temperature	-20℃~+60℃, 0%RH~80%RH	
output signal	UART (TTL)	Output voltage: 0~3.3V Input voltage: 0~3.3V compatible with 5V
	RS-485	ModBus-RTU communication protocol
	Analog output	Output voltage: 0~3V corresponds to 30~120dB
UART or RS-485 communication parameters	9600 N 8 1	
Measuring range	30dB~130dB	

Frequency weighting	A weighting																																					
Frequency response range	20Hz~12.5kHz																																					
Response time	Quick mode	500ms																																				
	Slow mode	1.5S																																				
stability	Less than 2% in use period																																					
Reference calibration point	94dB and 114dB calibration, reference sound pressure 20uPa, frequency 1kHz																																					
Noise accuracy	±0.5dB (in reference pitch, 94dB@1kHz)																																					
Dust cover effect	Influence ≤0.5dB in the range of 50~115db ≤0.7dB in other ranges within the range																																					
working environment	Air temperature: -40~+60° C Relative humidity: 25%~90% Static pressure: 65kPa~106kPa There are no strong mechanical vibrations, shocks, strong electromagnetic fields and corrosive gases around																																					
Adjustment data of sound pressure response and free field response	The equivalent free-field response can be obtained from the adjustment data in the table below by the sound pressure response generated by the acoustic calibrator or the simulated sound pressure response generated by the electrostatic actuator. <table><tr><th>频率 / kHz</th><th>自由场增量 / dB</th><th>频率 / kHz</th><th>自由场增量 / dB</th></tr><tr><td>1</td><td>0.2</td><td>6.3</td><td>2.2</td></tr><tr><td>1.25</td><td>0.3</td><td>8</td><td>3.4</td></tr><tr><td>1.6</td><td>0.4</td><td>10</td><td>5.0</td></tr><tr><td>2</td><td>0.5</td><td>12.5</td><td>6.2</td></tr><tr><td>2.5</td><td>0.7</td><td>16</td><td>7.6</td></tr><tr><td>3.15</td><td>0.9</td><td>18</td><td>8.4</td></tr><tr><td>4</td><td>1.3</td><td>20</td><td>9.0</td></tr><tr><td>5</td><td>1.8</td><td>—</td><td>—</td></tr></table>		频率 / kHz	自由场增量 / dB	频率 / kHz	自由场增量 / dB	1	0.2	6.3	2.2	1.25	0.3	8	3.4	1.6	0.4	10	5.0	2	0.5	12.5	6.2	2.5	0.7	16	7.6	3.15	0.9	18	8.4	4	1.3	20	9.0	5	1.8	—	—
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Directivity of the module at 1 kHz, 2 kHz, 4 kHz, 8 kHz																																						

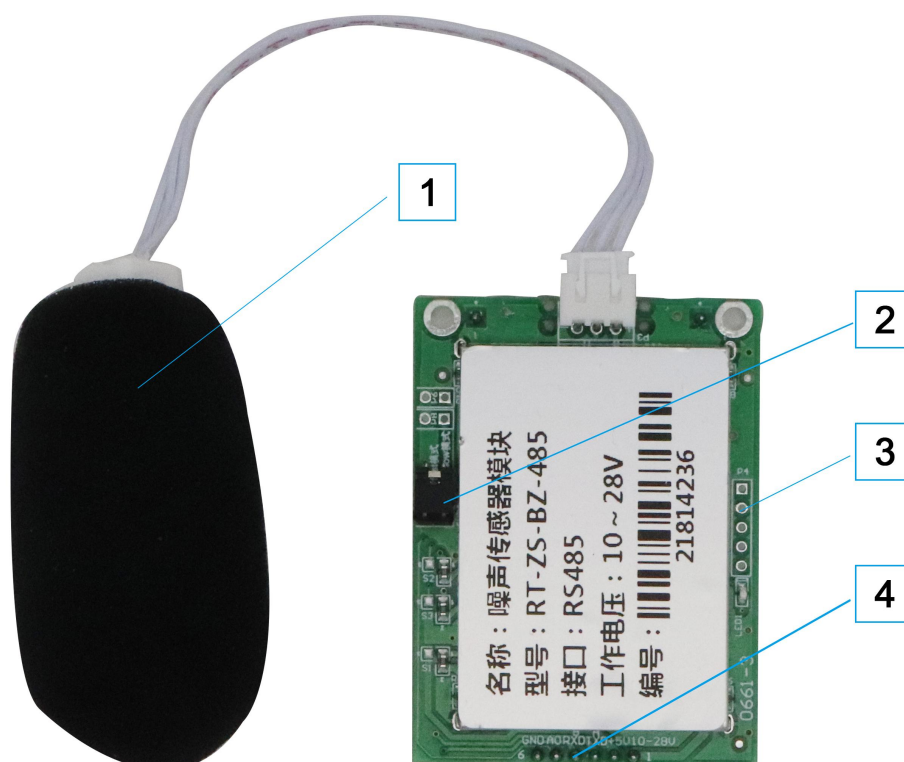


1.4product model

RT-					Company code
	ZS-				Noise module
		BZ-			PCB installation method
			TTL-		UART (TTL) output
			485-		485 output
				05	The supply voltage is 4.5~5.5V
				12	The supply voltage is 10~28V

2. Hardware description

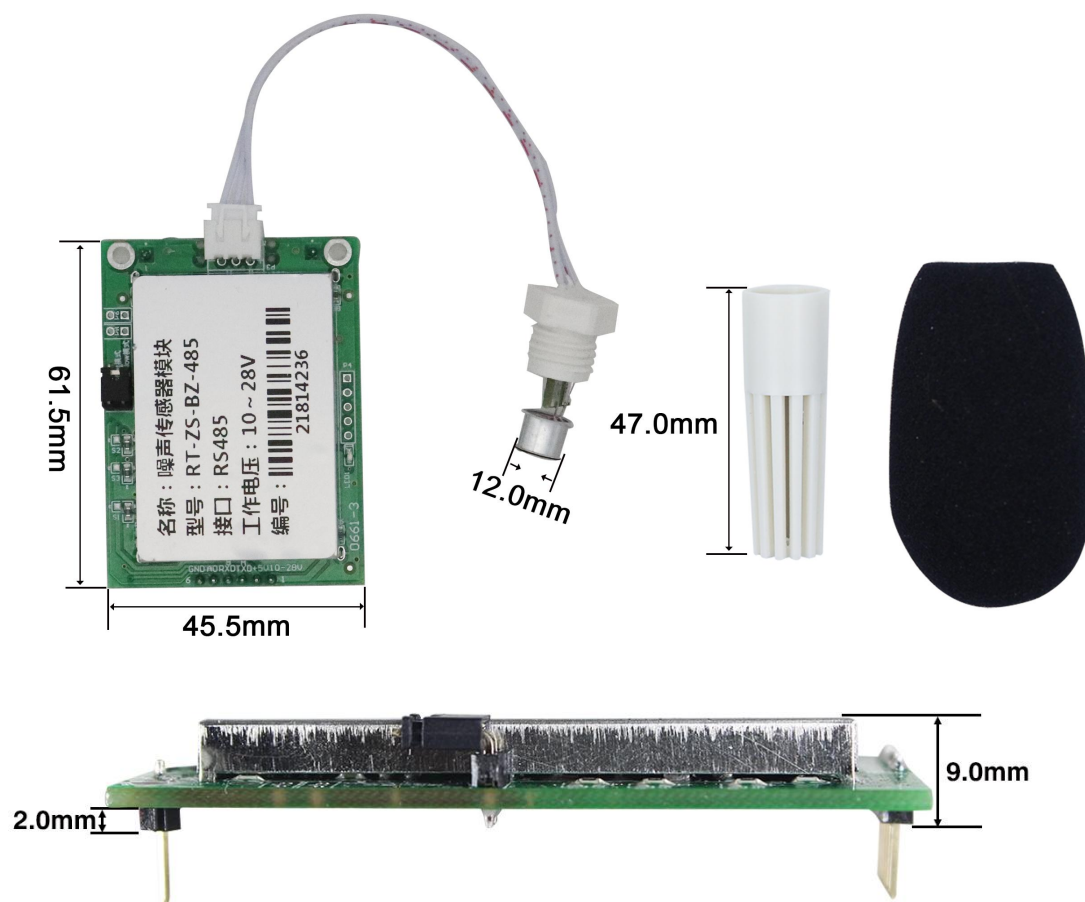
2.1 Device definition



Serial number	name		Description
1	Sound probe		Contains electret microphone, plastic jacket, dustproof cotton
2	Mode selection terminal		The jumper cap is shorted for fast mode, the noise update time is 500ms; the short is connected to the slow mode, and the noise update time is 1.5S.
3	Running lights		Running indicator, 1S flashes once
4	Pin number	name	Description
	1	10-28V	10-28V power input
	2	+5V	5V power input
	3	TXD/A	Serial data transmission (output voltage 0~3.3V) 485-A line
	4	RXD/B	Serial data reception (input voltage 0~3.3V tolerate 5V input) 485-B line
	5	A0	Analog signal output 0~3V

	6	GND	Negative power supply
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2.2 Equipment size

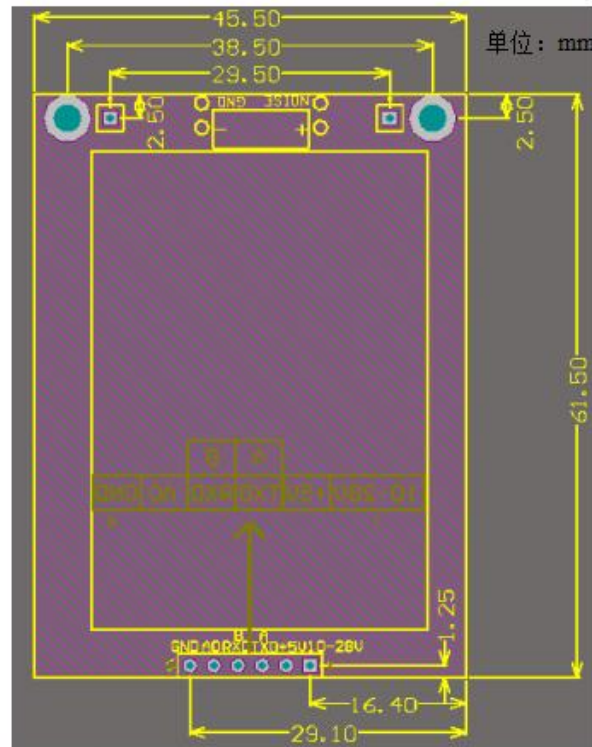


Note: The sound probe lead length is 10cm. To clamp the sound probe, the recommended panel opening size is $\phi 13\text{mm}$.

2.3 Module PCB package

The size chart is as follows:

Please download the package for Altium Designer package files.



3. Equipment communication

3.1 Communication basic parameters

The UART interface or 485 interface uses ModBus-RTU protocol for communication. The default communication communication parameters are:

Baud rate: 9600 Data bits: 8 bits Parity mode: No parity Stop bit length: 1 bit

The device ModBus communication address is

1.3.2 Data frame format definition

Adopt Modbus-RTU communication protocol, the format is as follows:

Initial structure $\geq$ 4 bytes of time

Address code = 1 byte

Function code = 1 byte

Data area = N bytes

Error check = 16-bit CRC code

End structure $\geq$ 4 bytes of time

Address code: default 01.

Function code: The instruction function of the command sent by the host. This module only uses function code 0x03 (read register data).

Data area: The data area is the specific communication data. Note that the 16-bit data high byte is

in front!

CRC code: Two-byte check code.

Host inquiry frame structure:

address code	function code	Register start address	Register length	Check code low	Check code high
1 byte	1 byte	2 bytes	2 bytes	1 byte	1 byte

Module response frame structure:

address code	function code	Effective number of bytes	Data area	Second data area	Nth data area	Check code
1 byte	1 byte	1 byte	2 bytes	2 bytes	2 bytes	2 bytes

3.3 Register address

Register address	PLC or configuration address	content	operating
0000 H	40001	Instantaneous noise value Upload data is 10 times true	Read only

3.4 Communication protocol example and explanation

Example: Read module noise value

Inquiry frame:

address code	function code	starting address	Data length	Check code low	Check code high
0x01	0x03	0x00 0x00	0x00 0x01	0x84	0x0A

Response frame: (for example, the current noise is 71.3dB)

address code	function code	Returns the number of valid bytes	Current noise value	Check code low	Check code high
0x01	0x03	0x02	0x02 0xC9	0x79	0x72

Noise calculation:

Current noise: 02C9H (hexadecimal) = 713 => noise = 71.3dB

4. Contact information

Shandong Renke Control Technology Co., Ltd.

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Post code: 250101

Phone: 400-085-5807

Website: www.renkeer.com

Cloud platform address: en.0531yun.cn Or: eniot.0531yun.cn

Web QR:



APP QR:



Android

4. Document history

The V1.0 documentation was created.

V2.0 updates product images